Lathi 20题练习册 第5批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

冲激乘积的化简与极限

Lathi 1.4-5 | 原书第 141、142、87 页

Boulet 第1章：Generalized Functions, p.26

(a)-(f)完整附上。提示所引式(1.10)已附；该式要求相乘函数在冲激位置连续，(f)按原提示用极限理解。

1.4-5 Simplify the following expressions:

(a) $\left(\frac{\sin t}{t^2+2}\right)\delta(t)$

(b) $\left(\frac{j\omega+2}{\omega^2+9}\right)\delta(\omega)$

(c) $[e^{-t}\cos(3t-60^\circ)]\delta(t)$

(d) $\left(\frac{\sin\left[\frac{\pi}{2}(t-2)\right]}{t^2+4}\right)\delta(1-t)$

(e) $\left(\frac{1}{j\omega+2}\right)\delta(\omega+3)$

(f) $\left(\frac{\sin k\omega}{\omega}\right)\delta(\omega)$

[*Hint:* Use Eq. (1.10). For part (f) use L'Hôpital's rule.]

$$\phi(t)\delta(t-T)=\phi(T)\delta(t-T) \qquad (1.10)$$

演算区

连续基本信号的奇偶分解与作图

Lathi 1.5-1 | 原书第 142 页

Boulet 第1章：Even and Odd Signals, p.23

1.5-1 Find and sketch the odd and the even components of the following:

(a) $u(t)$

(b) $tu(t)$

(c) $\sin \omega_0 t$

(d) $\cos \omega_0 t$

(e) $\cos(\omega_0 t + \theta)$

(f) $\sin \omega_0 tu(t)$

(g) $\cos \omega_0 tu(t)$

演算区

带饱和与延迟的系统性质判断

Lathi 1.7-4 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties, p.34

1.7-4 Input voltage $x(t)$ applied to an inverting op-amp follower circuit produces output $y(t)$ according to

$$y(t+t_{\mathrm{p}}) = \begin{cases} -V_{\mathrm{ref}} & x(t) > V_{\mathrm{ref}} \\ V_{\mathrm{ref}} & x(t) < -V_{\mathrm{ref}} \\ -x(t) & \text{otherwise} \end{cases}$$

where op-amp reference voltage V_{ref} and propagation delay t_{p} are both positive constants. Answer the following questions yes or no, and provide mathematical justification for each answer.

(a) Is this system BIBO-stable?
(b) Is the system causal?
(c) Is the system invertible?
(d) Is the system linear?
(e) Is the system memoryless?
(f) Is the system time invariant?

演算区

离散序列的时间变换、周期复制与能量功率

Lathi 3.2-6 | 原书第 316 页

Boulet 第1章：Transformations of the Time Variable，p.4；Periodic Signals，p.8；Finite-Energy and Finite-Power Signals，p.21

按原书离散时间扩展约定，扩展后下标非整数的位置补零。

3.2-6 Let DT signal $x[n]$ have values $[1, 2, 3, 4, 5, 6]$ for $0 \leq n \leq 5$ and let DT signal $y[n]$ have values $[5, 0, 0, 3, 0, 0, 1]$ for $0 \leq n \leq 6$. Both signals are zero outside the ranges given. Further, define a 6-periodic replication of $y[n]$ as $\tilde{y}[n] = \sum_{k=-\infty}^{\infty} y[n - 6k]$.

(a) Determine the energy E_x and power P_x of signal $x[n]$.

(b) Determine the smallest-magnitude integers N_1, N_2, and N_3 such that $y[n] = x[\frac{N_1 n}{N_2} + N_3]$.

(c) Determine the energy $E_{\tilde{y}}$ and power $P_{\tilde{y}}$ of $\tilde{y}[n]$.

演算区

离散冲激、阶跃与指数正弦恒等式

Lathi 3.3-2 | 原书第 316 页

Boulet 第1章：Discrete-Time Impulse and Step Signals， p.25；Periodic Complex Exponential and Sinusoidal Signals， p.17

3.3-2 Show that

(a) $\delta[n]+\delta[n-1]=u[n]-u[n-2]$

(b) $2^{n-1}\sin\left(\frac{\pi n}{3}\right)u[n]=\frac{1}{2}2^n\sin\left(\frac{\pi n}{3}\right)u[n-1]$

(c) $n(n-1)\gamma^n u[n]=n(n-1)\gamma^n u[n-2]$

(d) $(u[n]+(-1)^n u[n])\sin\left(\frac{\pi n}{2}\right)=0$ for all n

(e) $(u[n]+(-1)^{n+1}u[n])\cos\left(\frac{\pi n}{2}\right)=0$ for all n

演算区

由延迟冲激响应图判断因果性并卷积

Lathi 2.4-6 | 原书第 226 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

展开原题对2.4-5的引用：(a)判断系统是否因果并用数学说明；(b)用图解卷积求零状态响应 y_zsr(t)=x(t)*h(t)，反转并平移 x(t)，写出所有积分步骤并准确作图。本题图示为 h(t-1)，不是 h(t)。

2.4-6 Repeat Prob. 2.4-5 using the signals of Fig. P2.4-6, rather than those of Fig. P2.4-5. Be careful! Figure P2.4-6 shows $h(t-1)$, not $h(t)$.

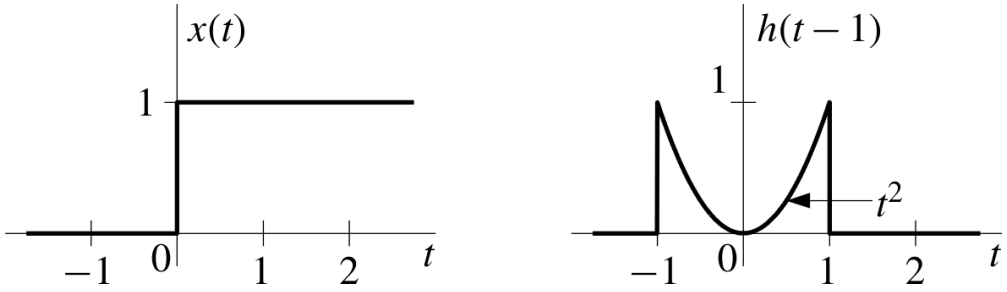


Figure P2.4-6

演算区

分段常数信号的图解卷积

Lathi 2.4-7 | 原书第 226 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

原图 P2.4-7 中的 x(t)、h(t) 均已附。

2.4-7 Suppose an LTIC system has impulse response $h(t)$ and input $x(t)$, both shown in Fig. P2.4-7. Use the graphical convolution procedure to determine $y_{zsr}(t) = x(t) * h(t)$. Accurately sketch $y_{zsr}(t)$. When solving for $y_{zsr}(t)$, flip and shift $x(t)$ and explicitly show all integration steps.

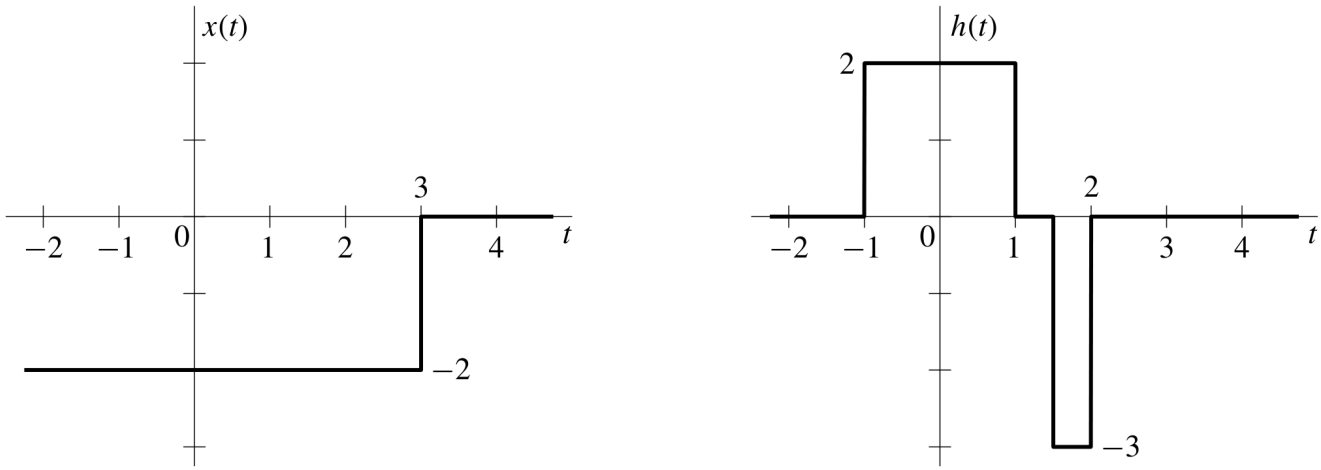


Figure P2.4-7

演算区

有限正弦脉冲与矩形脉冲的卷积

Lathi 2.4-26 | 原书第 228 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

所需原图 P2.4-26 的两个信号已完整附上。

2.4-26 Find and sketch $c(t) = x(t) * g(t)$ for the functions depicted in Fig. P2.4-26.

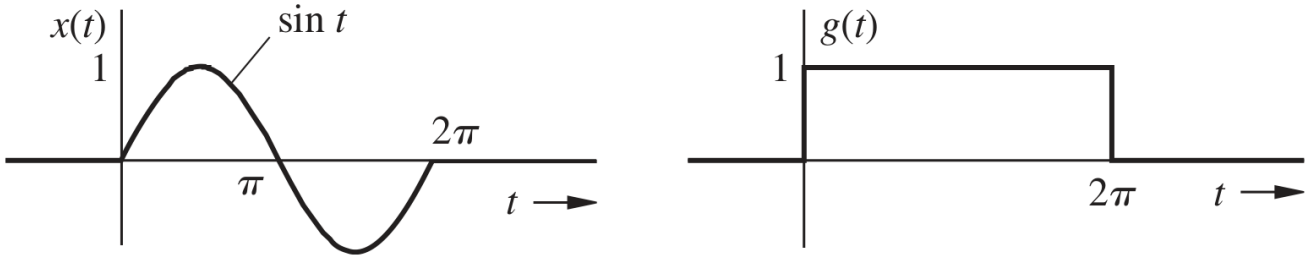


Figure P2.4-26

演算区

时移指数与带多项式因子的离散卷积

Lathi 3.8-8 | 原书第 322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， p.54

3.8-8 Find the (zero-state) response $y[n]$ of an LTID system if the input $x[n] = (3)^{-n+2}u[n+3]$, and

$$h[n] = 3(n-2)(2)^{n-3}u[n-4]$$

演算区

双边离散冲激响应的因果性、稳定性与卷积输出

Lathi 3.9-6 | 原书第 327 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

3.9-6 An LTID system has an impulse response given by

$$h[n] = \left(\tfrac{1}{2}\right)^{|n|}$$

(a) Is the system causal? Justify your answer.
(b) Compute $\sum_{n=-\infty}^{\infty} |h[n]|$. Is this system BIBO-stable?
(c) Compute the energy and power of input signal $x[n] = 3u[n-5]$.
(d) Using input $x[n] = 3u[n-5]$, determine the zero-state response of this system at time $n = 10$. That is, determine $y_{zsr}[10]$.

演算区

二阶系统特征模态与初始条件响应

Lathi 2.2-5 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按本题给出的初值求 t>=0 的零输入响应。

2.2-5 Repeat Prob. 2.2-4 for

$$(D^2+4D+4)y(t)=Dx(t)$$

and $y_0(0^-)=3,\ \dot{y}_0(0^-)=-4.$

演算区

连续系统的BIBO与内部稳定性分类

Lathi 2.5-2 | 原书第 233 页

Boulet 第3章：Characteristic Polynomials and Stability of Differential and Difference Systems, p.112

展开所引用2.5-1的要求：说明各系统在BIBO意义下稳定或不稳定，并判断其渐近稳定、不稳定或临界稳定；假设系统均可控且可观。

2.5-2 Repeat Prob. 2.5-1 for the following:

(a) $(D+1)(D^2+2D+5)^2y(t)=x(t)$

(b) $(D+1)(D^2+9)y(t)=(2D+9)x(t)$

(c) $(D+1)(D^2+9)^2y(t)=(2D+9)x(t)$

(d) $(D^2+1)(D^2+4)(D^2+9)y(t)=3Dx(t)$

演算区

二阶差分方程的初值递推

Lathi 3.5-3 | 原书第 319 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations, p.96

3.5-3 Solve the following equation recursively (first three terms only):

$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 0$$

with

$$y[-1] = -25,\ y[-2] = 0.$$

演算区

非相邻辅助条件下的离散零输入响应

Lathi 3.6-8 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

注意：原题两个辅助条件分别给在 n=-1 和 n=1。

3.6-8 Consider the discrete-time system $y[n]+y[n-1]+0.25y[n-2]=\sqrt{3}x[n-8]$. Find the zero input response, $y_0[n]$, if $y_0[-1]=1$ and $y_0[1]=1$.

演算区

重特征根离散系统的单位冲激响应

Lathi 3.7-5 | 原书第 321 页

Boulet 第3章：Impulse Response of a Difference LTI System， p.109

展开所引用3.7-1的要求：求方程所描述系统的单位冲激响应 h[n]。

3.7-5 Repeat Prob. 3.7-1 for

$$(E^2 - 6E + 9)y[n] = Ex[n]$$

演算区

半波对称性、奇次谐波与周期信号级数

Lathi 6.1-6 | 原书第 669、671 页

Boulet 第4章：Properties of Continuous-Time Fourier Series， p.139；Determination of the Fourier Series Representation， p.134

半波对称命题、两个待证系数公式和原图 P6.1-6(a)、(b)均已附；系数公式是原题要求证明的命题。

6.1-6 If the two halves of one period of a periodic signal are identical in shape except that one is the negative of the other, the periodic signal is said to have a *half-wave symmetry*. If a periodic signal $x(t)$ with a period T_0 satisfies the half-wave symmetry condition, then

$$x\left(t-\frac{T_0}{2}\right)=-x(t)$$

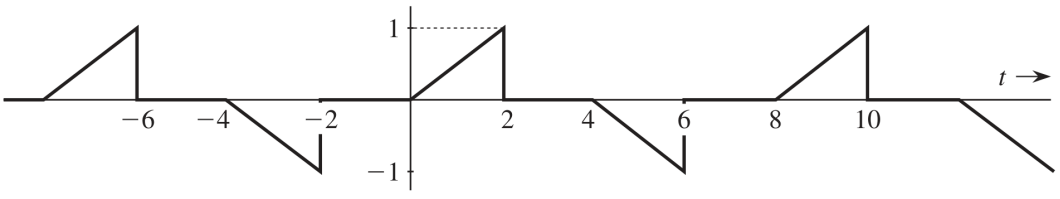
In this case, show that all the even-numbered harmonics vanish and that the odd-numbered harmonic coefficients are given by

$$a_n=\frac{4}{T_0}\int_0^{T_0/2}x(t)\cos n\omega_0 t\,dt$$

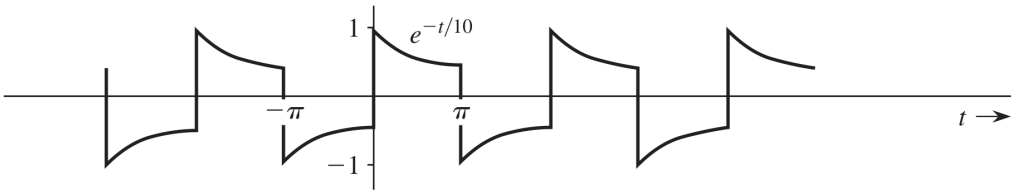
and

$$b_n=\frac{4}{T_0}\int_0^{T_0/2}x(t)\sin n\omega_0 t\,dt$$

Using these results, find the Fourier series for the periodic signals in Fig. P6.1-6.



(a)



(b)

Figure P6.1-6

由复指数幅相谱重建三角级数

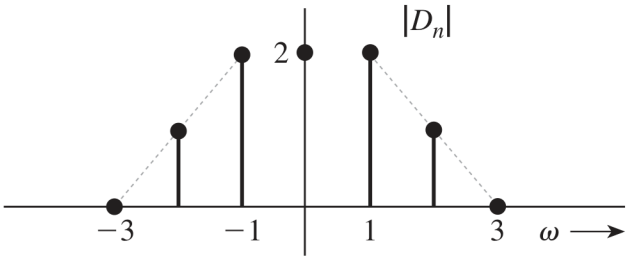
Lathi 6.3-7 | 原书第 673 页

Boulet 第4章：Graph of the Fourier Series Coefficients: The Line Spectrum， p.137

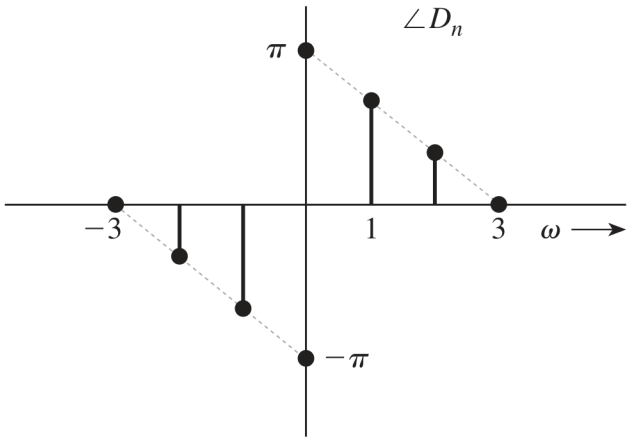
原题同页跨栏，(a)-(d)与原图 P6.3-7 的两幅频谱均已附。

6.3-7 Figure P6.3-7 shows the exponential Fourier spectra of a periodic signal $x(t)$.

(a) By inspection of Fig. P6.3-7, find the exponential Fourier series representing $x(t)$.
(b) By inspection of Fig. P6.3-7, sketch the trigonometric Fourier spectra for $x(t)$.
(c) By inspection of the trigonometric Fourier spectra found in part (b), find the trigonometric Fourier series for $x(t)$.
(d) Show that the series found in parts (a) and (c) are equivalent.



(a)



(b)

Figure P6.3-7

演算区

三角波级数及其时移、尺度变换

Lathi 6.3-9 | 原书第 673、674 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Properties of Continuous-Time Fourier Series，p.139

本题原书跨页；原图 P6.3-9(a)、(b)、(c)已完整附上。

6.3-9 (a) Find the exponential Fourier series for the signal in Fig. P6.3-9a.

(b) Using the results in part (a), find the Fourier series for the signal $\hat{x}(t)$ in Fig. P6.3-9b, which is a time-shifted version of the signal $x(t)$.

(c) Using the results in part (a), find the Fourier series for the signal $\tilde{x}(t)$ in Fig. P6.3-9c, which is a time-scaled version of the signal $x(t)$.

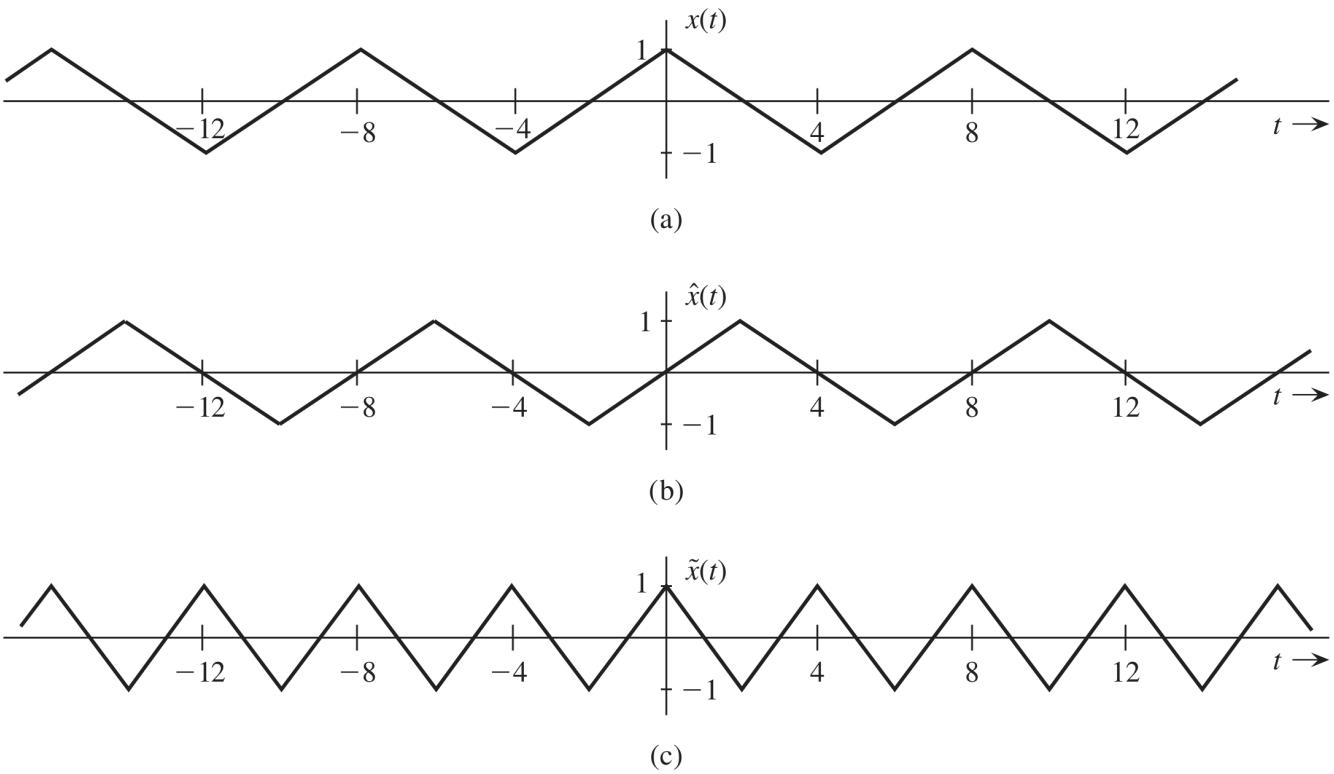


Figure P6.3-9

给定传递函数的周期指数输入响应

Lathi 6.4-1 | 原书第 675、599 页

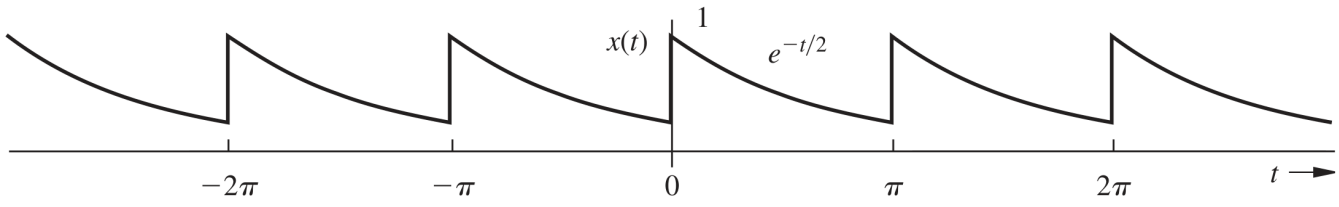
Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal，p.155

所需输入原图6.2(a)已附。该图的一个周期为从 t=0 到 t=pi 的 exp(-t/2) 段。

6.4-1 Find the response of an LTIC system with transfer function

$$H(s) = \frac{s}{s^2 + 2s + 3}$$

to the periodic input shown in Fig. 6.2a.



(a)

演算区

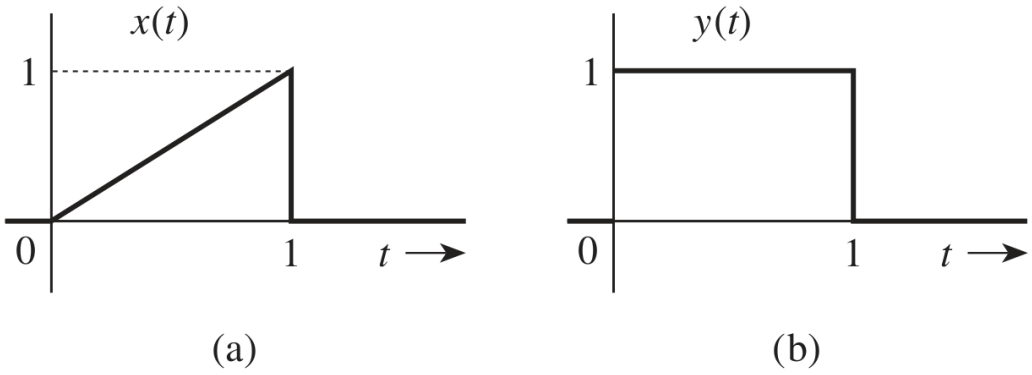
最小误差能量下的反向信号投影

Lathi 6.5-5 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

所需原图 P6.5-4 的 x(t)、y(t)均已附。本题用 c*x(t) 逼近 y(t)。

6.5-5 For the signals $x(t)$ and $y(t)$ shown in Fig. P6.5-4, find the component of the form $x(t)$ contained in $y(t)$. In other words, find the optimum value of c in the approximation $y(t) \approx cx(t)$ so that the error signal energy is minimum. What is the error signal energy?



(a) (b)

Figure P6.5-4

演算区